

NON-TECHNICAL OVERVIEW · SEPTEMBER 2026

Rock dust for farmers and for the climate: where it pays in Sub-Saharan Africa

A map-based study of where spreading crushed basalt on cropland would pay for itself through better yields, through carbon removal, or **only when the two are added together**, and how the model behind it has matured.

44

countries mapped, patch by patch

23

crops, weighted by where they are grown

~9 km

size of each patch in the map

2.7 Mha

cropland that pays only when yield and carbon are combined

AUTHORS

B. H. Gebrekidan (CIMMYT-Ethiopia) · J. Chamberlin (CIMMYT-Kenya)

WHAT THIS IS

Companion to the technical walkthrough, without the equations

BASIS

Model run of 1 September 2026

STATUS

Working draft · manuscript in preparation

THE IDEA

One field practice, two benefits.

WHAT IS DONE

Basalt, a common volcanic rock, is crushed to a fine dust and spread on fields with the same equipment used for agricultural lime.

BENEFIT TO THE FARMER

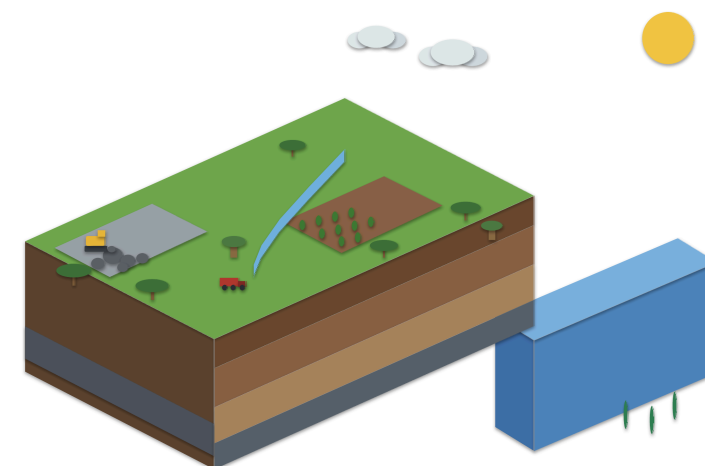
As the dust dissolves it neutralizes soil acidity, exactly as lime does. Acid soils are a major reason yields stay low, so treating them raises harvests.

BENEFIT TO THE CLIMATE

The same reaction pulls carbon dioxide out of the air and turns it into dissolved bicarbonate, which drains to rivers and the ocean and stays there for thousands of years.

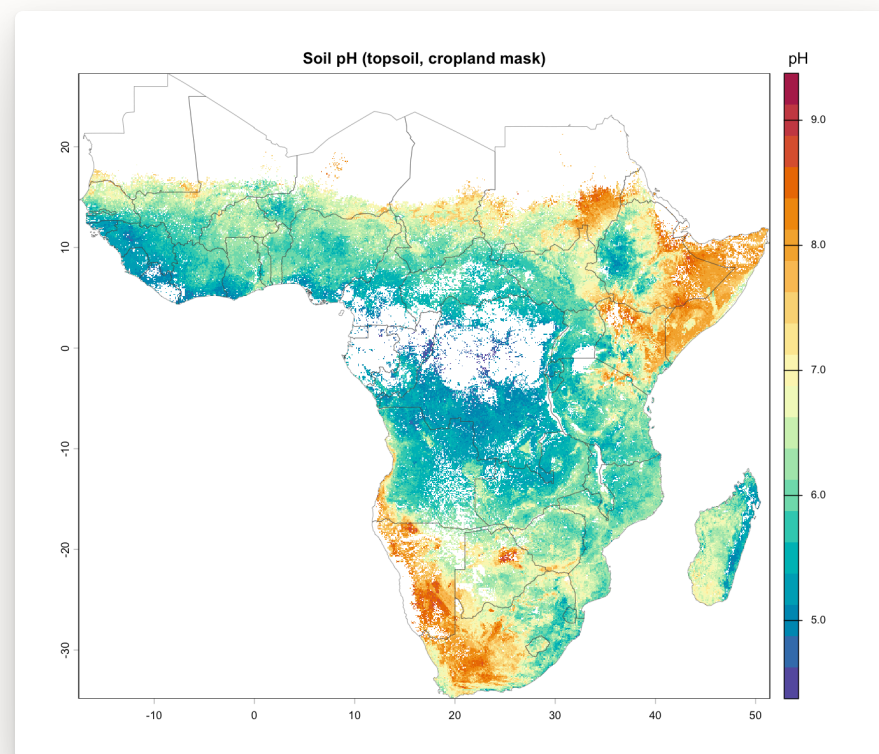
WHY IT IS DIFFERENT

Lime also fixes acidity but releases its own carbon. Basalt fixes acidity and removes carbon. That second benefit can be sold to carbon-credit buyers.



WHY SUB-SAHARAN AFRICA

Acid soils are widespread, and rarely treated.



Blue and green on the map are acid soils. They cover the humid belt from West Africa through Central Africa to the highlands of the east.

~23%

of cropland is acid enough to depress yields (about 33 million hectares)

~\$6_B

yearly value of harvest lost to soil acidity, by recent estimates

Liming is rare because the upfront cost is large and smallholders cannot easily borrow against a benefit that arrives over several seasons. Warm, wet conditions also make basalt dissolve faster here than in temperate regions.

THE QUESTION

Where would it pay, and who would pay for it?

For every 9 km patch of cropland we ask the same three questions, each against the full cost of buying, grinding, hauling and spreading the rock.

THE FARMER'S CASE

Does the extra harvest alone cover the cost?

If yes, farmers or input dealers could fund it themselves and no carbon money is needed.

THE CARBON BUYER'S CASE

Does the carbon removed alone cover the cost?

If yes, a carbon-credit buyer could fund it purely as climate action, with the yield gain as a bonus.

THE COMBINED CASE

Does it pay only when both are added?

If yes, carbon money is the missing piece that turns an unaffordable farm investment into a viable one.

Why the split matters. The three answers point to three different financing routes: private input markets, carbon-credit purchases, and a blend where the credit acts like a cost-share. The map tells us which route fits which place.

HOW THE MODEL WORKS

Three streams add up to one balance per patch.

$$\text{Return per hectare} = \text{value of extra harvest} + \text{value of carbon removed} - \text{delivered cost of rock}$$

HARVEST STREAM

How acid is the soil, which crops grow there, how much each crop responds to acidity relief, and what the crop sells for locally.

CARBON STREAM

How much carbon a tonne of rock removes in that climate and soil, minus carbon that never leaves the field, valued at a carbon price net of monitoring cost.

COST STREAM

Rock at the quarry, electricity to grind it, truck time from the nearest basalt source over real roads, and spreading on the field.

TWO WAYS TO DOSE

Targeted: just enough rock to fix each soil's acidity, like a lime recommendation. **Uniform:** a flat 10, 20 or 50 tonnes per hectare everywhere, the convention in the carbon literature.

TWO MOMENTS IN TIME

First application: a big one-off dose. **Steady state:** a small yearly top-up once soils are fixed. Headline results use the steady state, where both benefits recur every year.

WHAT COMES OUT

A map of the balance for every patch under four doses and three timings, a ranking of the best patches under yearly rock-supply limits, and a 180-run stress test over prices, costs, yields and carbon rates.

WHAT GOES IN

Every input is a public dataset.

WE NEED TO KNOW	WHERE IT COMES FROM	IN PLAIN TERMS
How acid the soil is	SoilGrids	Global soil maps: pH, acidity held on soil particles, texture
What grows where	SPAM	Maps of 23 crops: area, production and yield
How crops respond to pH	EcoCrop	Each crop’s comfortable pH range and how yield falls outside it
Where basalt rock is	GLiM geology map	Volcanic rock outcrops that could supply quarries
How long a truck takes	Malaria Atlas friction surface	Travel time over the actual road network from rock to field
How fast rock dissolves	WorldClim	Temperature and rainfall, which speed up or slow down weathering
What crops are worth	FAOSTAT	Farm-gate prices, 2016 to 2020 medians
How much rock each soil needs	LiTAS lime calculator	The lime-equivalent dose, translated to basalt

Everything is reproducible from the project repository; large input maps are shared on request. Constants such as the quarry price, the trucking rate and the carbon price are all adjustable.

COST

The truck, not the quarry, sets the price.

\$10/t

rock at the quarry gate

~\$41/t

average delivered to the field, before grinding and spreading

~55%

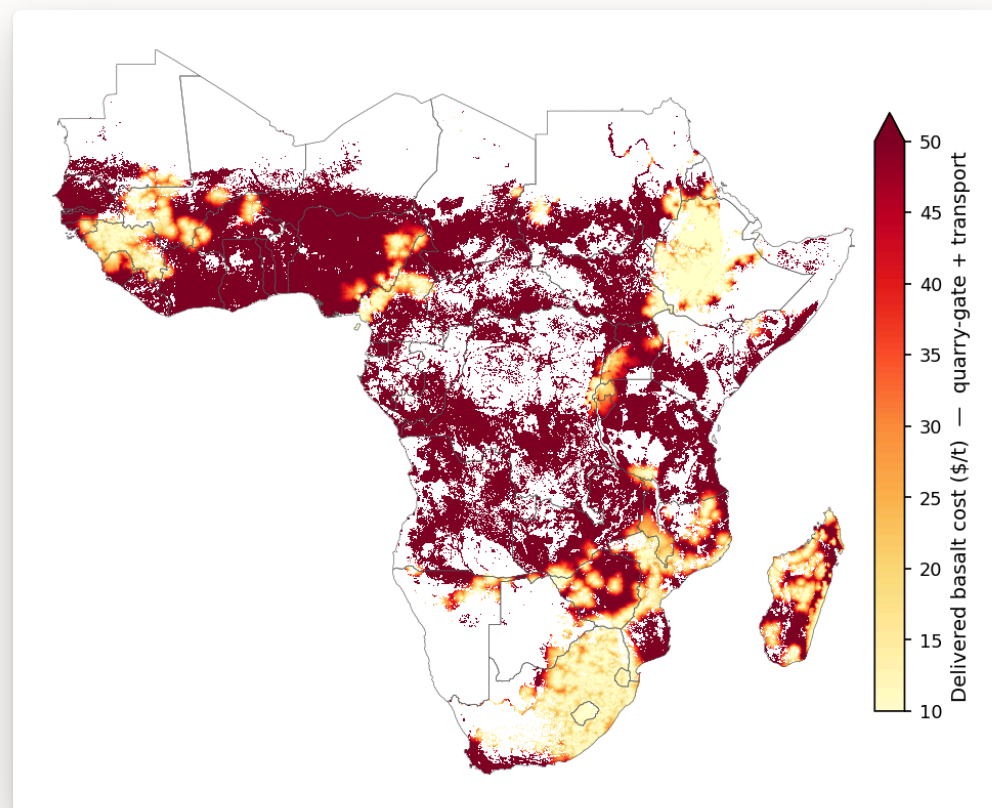
share of delivered cost that is transport, on average

READING THE MAP

Pale areas sit near basalt sources and get rock cheaply. Dark red areas are far from any source: even with excellent soils and crops, rock there costs too much to move.

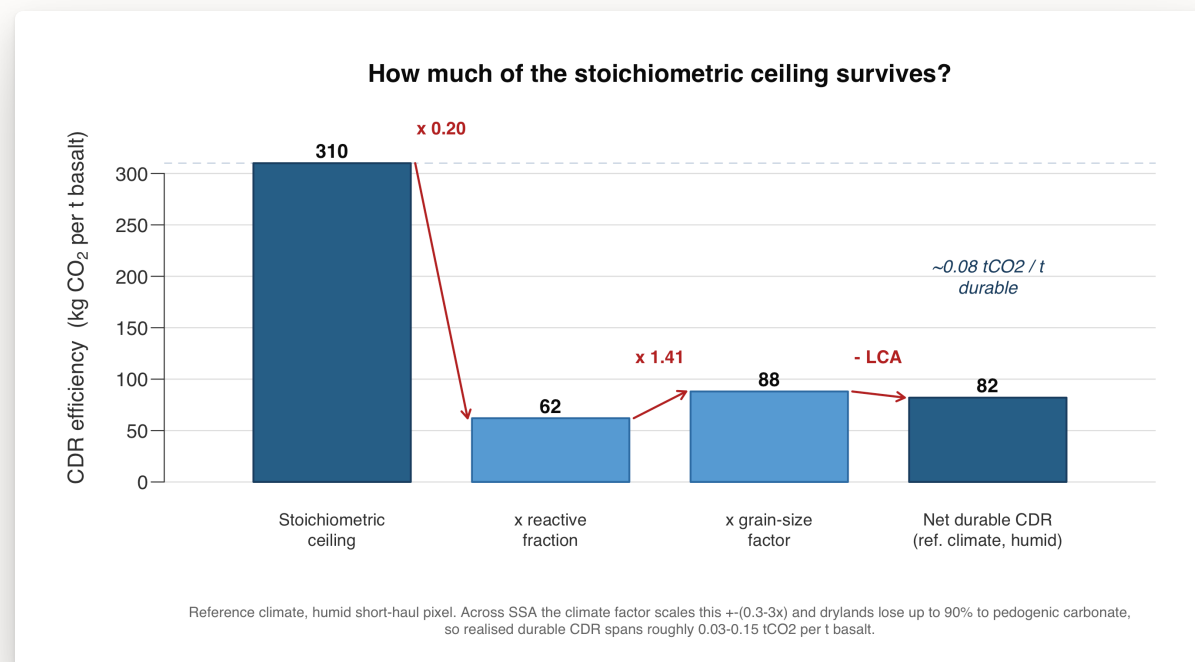
WHY THIS MATTERS

Transport is the one cost that varies from place to place, so it decides *where* the practice is possible. Cheaper haulage, for example electrified trucking, is a real lever.



CARBON

A tonne of rock removes far less than the textbook number.



In theory a tonne of basalt can lock up about 310 kg of carbon dioxide. In a real field, only part of the rock dissolves within a useful time, and hauling and grinding emit some carbon of their own.

WHAT WE USE

Roughly **80 to 90 kg** per tonne in a warm, humid, well-connected place, scaled up or down by local temperature, rainfall and soil acidity. Dry areas lose more because carbon can re-precipitate in the soil.

THE BIGGEST UNKNOWN

This number is the least-measured quantity in the whole field, especially for tropical smallholder soils. Every result about carbon should be read with that in mind.

CARBON · AN ACCOUNTING RULE WE ADDED THIS YEAR

Fixing acidity uses up carbon before any is stored.

THE MECHANISM

The alkalinity that basalt releases can either neutralize soil acidity or drain away as stored carbon. It cannot do both. The part that neutralizes acidity gives its carbon back to the air, exactly as lime does.

THE RULE

We credit only the carbon that is left over after the soil's acidity has been paid down. Most global models skip this step; the 2026 systematic review of the field notes it is “not considered in all global and regional model estimates.”

TWO CONSEQUENCES

First, the dose that is best for the farmer (just enough to fix the soil) removes almost no carbon. Second, durable removal is mostly a steady-state story: at first application under a third survives, in steady state most of it does.

~70%

of credited carbon disappears at first application once acidity is netted out

77–90%

survives in steady state, when the yearly top-up only offsets fresh acidity

What it means for buyers. The first application is, in carbon terms, mostly liming. The carbon case belongs to programs that keep fields treated year after year.

PART I

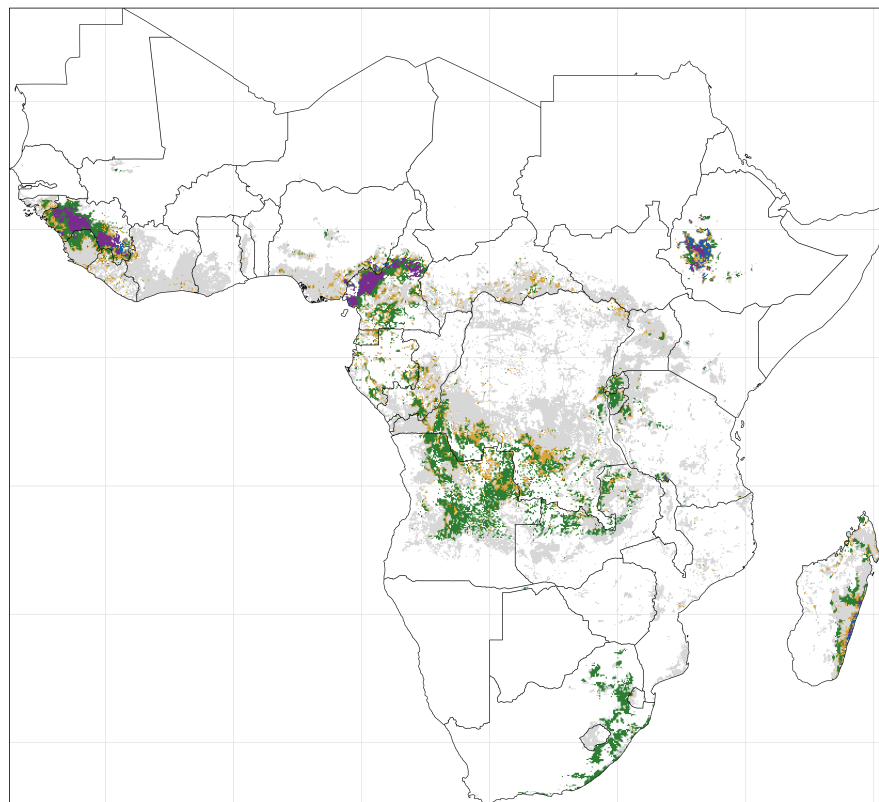
What we found

Where yield alone pays, where carbon alone pays, where only the combination does, and where to start.

RESULTS • THE MAP

Every patch of cropland falls into one of five types.

Return typology • equilibrium, net-export CDR, targeted



both private only public only combined only neither

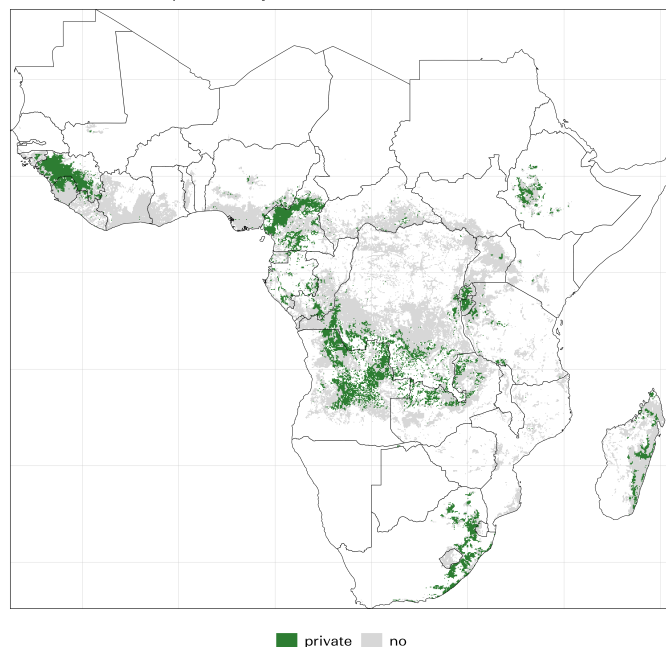
- **Yield pays on its own** • farmers could fund it
- **Carbon pays on its own** • a credit buyer could fund it
- **Both pay on their own** • doubly justified, but the credit adds least
- **Only together** • neither alone, but the sum covers cost
- **Neither** • not worth it at today's prices and costs

Targeted dose, steady state, \$150 per tonne of CO₂ net of \$20 monitoring.

RESULTS • THE FARMER'S CASE

Yield alone pays on about four million hectares.

Private-sufficient (yield) • equilibrium



Land where the extra harvest covers the full cost of the rock with no carbon payment at all.

4.3 Mha

cropland where yield alone pays

~2:1

value of extra harvest to cost of rock
per steady-state cycle (\$3.5 B
against \$1.7 B)

This is the largest single group, and it barely changes under any carbon assumption. But it is a minority of acid cropland: most viable land needs more than yield to pay.

RESULTS · THE CARBON BUYER’S CASE

Carbon alone pays on less land, and depends on the price.

At today’s conservative price of \$150 per tonne, durable carbon covers the cost on about **2 million hectares** with the targeted dose, and about **4 million** with a flat 20 t/ha dose that leaves more alkalinity free to drain away.

THE PRICE LADDER

Carbon-only viability is steeply price-dependent. Between \$250 and \$350 a large share of treated cropland tips over the line. Current buyers describe \$271 as expensive, and suppliers say they need about \$349 to be profitable.

WHAT IT COSTS TO REMOVE A TONNE

Across treated land the typical cost of one durably removed tonne is about **\$276** with the flat dose and \$330 with the targeted dose, against a net price of \$130 after monitoring. Only the cheapest patches clear today.

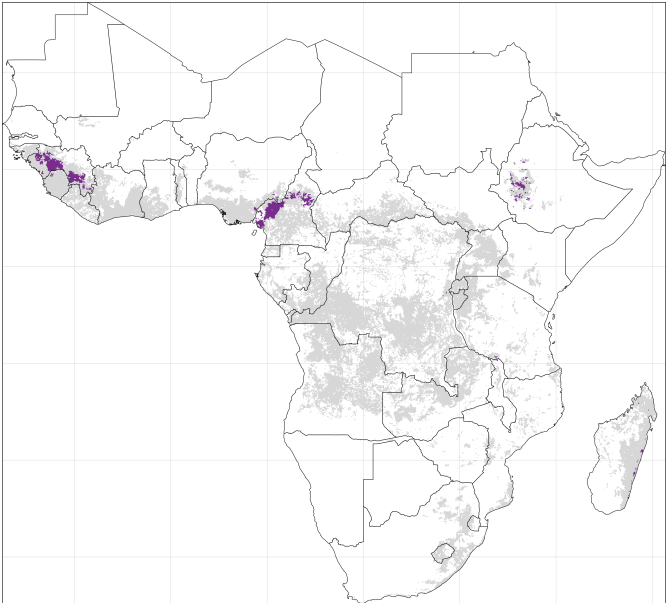
Carbon pays on its own · flat 20 t/ha dose, steady state

CARBON PRICE	AREA (MHA)	SHARE OF TREATED LAND	CO ₂ (MT)
\$100	1.4	4%	7.8
\$150	4.0	12%	20.4
\$250	7.0	22%	33.3
\$350	23.1	71%	100.6
\$500	28.0	86%	115.6

RESULTS · THE HEADLINE

The largest opportunity is land where only the combination pays.

Intersection · equilibrium, net-export CDR



■ intersection ■ outside

4.3

Mha where yield alone pays

2.0

Mha where carbon alone pays

7.6

Mha where the combination pays

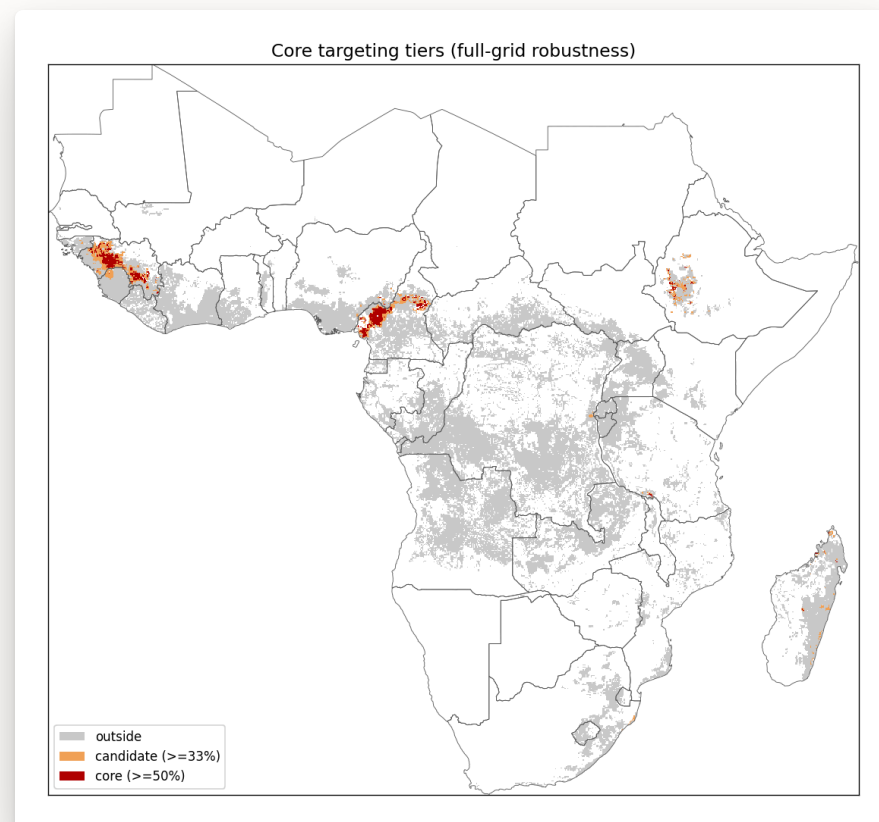
2.7 MILLION HECTARES

Neither the harvest nor the carbon covers the cost on its own, yet together they do. That is more than either single-benefit group.

In one sentence: carbon payments are not a bonus on a practice that already pays; on most viable land they are the missing margin that makes a yield-improving investment affordable. That also makes the credit clearly additional: without it, the practice does not happen.

RESULTS · WHERE TO START

A small core stays attractive under most assumptions.



We re-ran the map 180 times with different costs, prices, yield responses and carbon rates, and kept the land that stays viable in most of them.

0.65 Mha

robust core (viable in at least half the runs); candidate tier
1.22 Mha

0.38 M

farms in the core, order of magnitude

~\$0.6 B

crop value at stake in the core

The core sits in the **Cameroon highlands** and the **Guinea highlands**, with thinner extensions into Ethiopia, Tanzania and Sierra Leone: acid soils, warm wet weather, basalt nearby, and crops worth selling, all in one place. This is where pilots belong.

RESULTS · WHAT WOULD CHANGE THE PICTURE

Two levers move the carbon case; one moves the farmer's.

1 · CARBON REMOVED PER TONNE OF ROCK

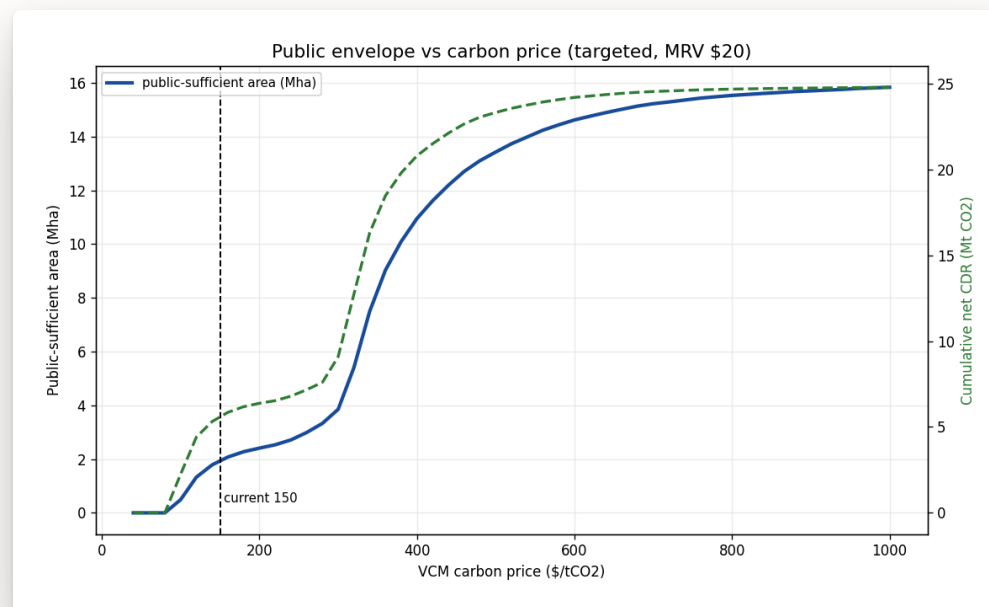
The strongest lever. If a tonne of rock verifiably removed twice as much carbon, the land where carbon alone pays would grow about **six-fold**. Better measurement raises this number as much as better rock does.

2 · CARBON PRICE RELATIVE TO DELIVERED COST

Price, monitoring cost and delivered cost act as one lever, through their ratio. A higher price or cheaper rock both help, but less than lever 1 for large moves.

3 · CHEAPER HAULAGE

Cutting transport cost by a quarter, for instance with solar-electric trucking, expands the farmer's case from 4.3 to 5.1 Mha but barely moves the carbon case. Logistics helps farmers; price and measurement help buyers.



PART II

Progress and next steps

How the model has matured since May 2026, how the numbers moved as it did, what it still leaves out, and what comes next.

PROGRESS · HOW THE MODEL MATURED

From framework to a fully re-anchored model in four months.

MAY 2026 · BUILD

Framework and first full run

Conceptual framework and pipeline figures (14 May). Road-network logistics wired in (15 May). First end-to-end profitability run across 44 countries and 23 crops (19 May). Conservatism review re-anchored five defaults (26 May).

JULY 2026 · ACCOUNTING

Net-export carbon rule

Credit only carbon that leaves the field after acidity is paid down. Carbon case re-led on the flat 20 t/ha dose. Layered sensitivity harness, public-envelope sensitivity, comparison with Cascade Climate’s cost tool. Manuscript drafted; coauthor deck.

1 SEPT 2026 · EVIDENCE

Literature audit and re-run

Every load-bearing citation checked against the published record, anchored to the 2026 systematic review of the field. Temperature response moved to the literature-standard form; soil-pH optimum moved from 5 to 6; yield sweep capped at field-trial evidence. All 276 output maps regenerated.

2 SEPT 2026 · NOW

Manuscript re-anchored

Results, tables and figures updated to the new basis. Section-level citations to the anchor review added throughout. Draft marked not for circulation. Technical walkthrough deck rebuilt; this overview added.

The nine building blocks of the model and where each stands

BLOCK 1

Grid

live

BLOCK 2

Dose

live

BLOCK 3

Carbon

re-anchored 1 Sept

BLOCK 4

Logistics

live

BLOCK 5

Energy

live

BLOCK 6

Crop response

live; trial-fitted variant not yet run

BLOCK 7

Profitability

re-run 1 Sept

BLOCK 8

Supply ranking

re-run 1 Sept

BLOCK 9

Sensitivity

re-run 1 Sept

PROGRESS · HOW THE NUMBERS MOVED

The carbon numbers moved; the yield numbers did not.

QUANTITY	JULY DRAFT	SEPTEMBER BASIS	WHY IT MOVED
Land where yield alone pays	4.3 Mha	4.3 Mha	Unchanged: the farmer’s case does not depend on carbon assumptions
Land where carbon alone pays	1.7 Mha	2.0 Mha (4.0 at flat 20 t/ha)	Warmer-climate weathering response; carbon case re-led on the flat dose
Land where only the combination pays	2.1 Mha	2.7 Mha	Same reasons; combined area 6.9 to 7.6 Mha
Cost of removing a tonne of CO ₂ (median)	\$446	\$276 (flat dose) / \$330 (targeted)	Now area-weighted; more carbon removed per tonne of rock
Robust core for pilots	0.8 Mha · 0.5 M farms	0.65 Mha · 0.38 M farms	Robustness now tested on the same basis as the headline, a stricter test
Carbon credited at first application	full weathering	about 30% of it	Net-export rule: acidity is paid down first

Our median removal cost now sits at the upper edge of the published range (\$137 to \$276 per tonne across 25 costing studies). The gap is accounting stringency: we credit only carbon that durably leaves the field, where most studies credit gross removal.

PROGRESS · WHAT THE 2026 SYSTEMATIC REVIEW TELLS US

The field agrees on the question we are asking.

NOT WHETHER, BUT WHERE

The review of 241 studies concludes that the question is no longer whether rock weathering works, but “where, when, and how it can be made to work most effectively.” That is exactly what a map answers.

THE MARKET IS SMALL

About 0.8 million tonnes of rock-weathering credits have been sold, 1 to 2% of all durable removal credits, mostly to two buyers. Few tonnes have been delivered yet.

YIELD BENEFITS ARE REAL

Across 166 agronomic studies, 129 report gains and 11 losses. Field trials in maize show 10 to 70% more harvest, and the response depends on relieving a soil constraint, not on piling on more rock.

POLICY POINTS THE SAME WAY

The review calls for subsidies “where on-farm benefits do not fully cover the cost” and for folding rock dust into existing liming subsidies. Our combined-only band is the map of where that condition holds.

CARBON NUMBERS ARE WIDE

Measured removal rates span several orders of magnitude, with a median of 0.84 tonnes per hectare per year, and tend to sit below what global projections assume.

AFRICA IS UNDER-STUDIED

Research is led from the UK, US and Germany; large parts of Africa are “sparsely represented,” even though the highest potential is in the tropics. Our model runs on the least-measured region.

Suhrhoff et al. (2026), An Ecosystem of Carbon Dioxide Removal Reviews, Part 3: Enhanced Weathering. CDRXIV preprint 417, submitted to Energy & Environmental Science, not yet peer reviewed.

CAVEATS

What the model does not yet do.

Who pays upfront

We map where returns cover costs, not who can finance the first dose. Tenure, credit and cost-sharing are not modelled.

Thin yield evidence

The crop response rests on a dozen or so trials plus a crop-suitability model. Reassuringly, the map is least sensitive to this input.

Carbon lost downstream

Some stored carbon escapes in rivers and at the ocean surface, roughly 15 to 20% in published estimates. We do not deduct it, so carbon results lean slightly high.

Carbon that arrives late

Soils can hold weathered material for years before it drains. We treat that as a conservative deduction; some of it will eventually count.

Other greenhouse gases

Most studies find rock dust reduces nitrous oxide from fields. We leave this out, which works against the practice, not for it.

Farm counts

Derived from national average holding sizes, so they are order-of-magnitude indications, not enumerations.

HOW TO READ THE RESULTS

The ordering of the findings is robust: yield pays on more land than carbon, the combined-only band is the largest opportunity, and the core geography holds across the stress test. The absolute carbon numbers move with the removal rate per tonne, the input we most need to measure.

NEXT STEPS

Measure the carbon, pilot the core, publish the map.

FIELD DATA, IN ORDER OF VALUE

1 • Carbon removed per tonne, in tropical smallholder soils

The single input that most moves both the carbon case and the priority map. Needs multi-year trials, not one-season ones.

2 • Carbon re-precipitation in semi-arid zones

Today a rule of thumb by aridity band; a measured fraction would tighten the drier half of the map.

3 • Delivered cost near candidate basalt sources

Real quarry and trucking quotes in the Cameroon and Guinea highlands.

4 • Yield response on African trials

Lowest leverage, since the map changes least with this input; new Ghana and Kenya trials feed straight in.

MODEL AND OUTPUTS

Registry-grade accounting

Align our carbon rule with the counterfactual and additionality tests that credit registries actually apply.

Run the trial-anchored yield model

Replace the crop-suitability fallback with the Bayesian fit on field trials, and widen the trial set as new African results appear.

Pilot design in the core

Put the first pilots where a measurement is most likely to change a decision, and monitor them for more than one season.

Manuscript and tools

Submit to Agricultural Economics. The interactive explorer already lets anyone move every lever of the analysis in a browser.

TAKEAWAYS

Four things to remember.

1

The two benefits rarely pay on their own

But together they make about 2.7 million hectares of acid cropland worth treating that neither would alone.

2

Carbon money is the missing margin, not a bonus

On most viable land the credit is what makes a yield-improving investment affordable. That is a cost-share for farmers, and a clearly additional credit for buyers.

3

The carbon case hinges on one number

How much CO₂ a tonne of rock verifiably removes. It is the least-measured quantity in the field and the first thing to go and measure.

4

There is a place to start

The Cameroon and Guinea highlands stay attractive under most assumptions. Pilots there would test the model where a result is most likely to change a decision.



DISCUSSION

Thank you for your interest!

Rock dust for farmers and for the climate:
where it pays in Sub-Saharan Africa

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CIMMYT · plain-language overview · working draft, September 2026

